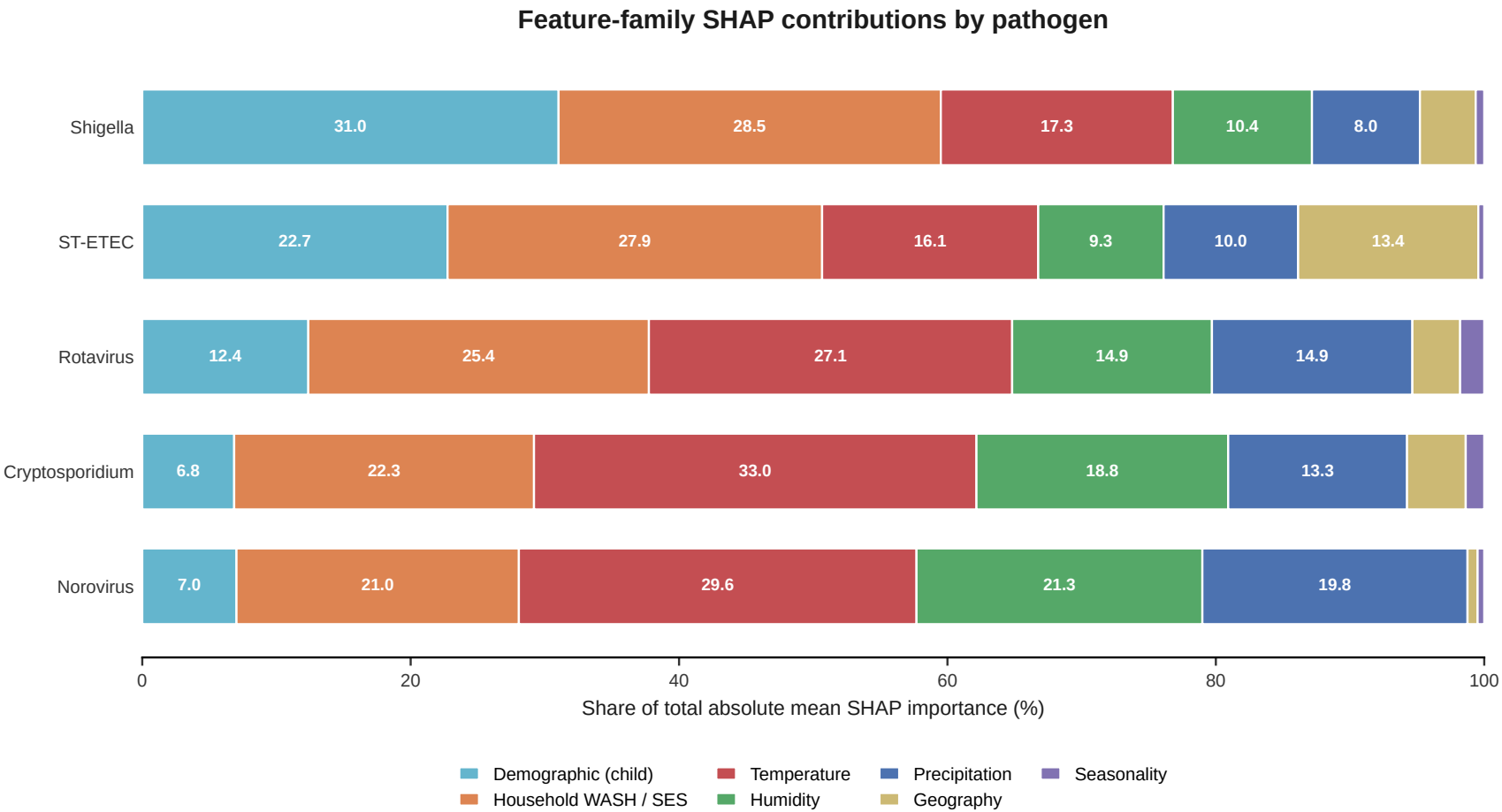

Contributions are from CatBoost surrogate models refitted to the full cleaned analytic datasets, expressed as percentages of the total absolute mean SHAP importance within each pathogen. The seven families were defined after initial inspection of the fitted models rather than a priori.